

Le Nguyen Thanh Ha

AI Product Engineer — I turn operational problems into revenue-generating software, and I ship it end to end.

+84 396 913 492 • thanhha.le@outlook.com • linkedin.com/in/hale03 • github.com/miniHaLe • Ho Chi Minh City — open to relocation / remote

EXECUTIVE SUMMARY

I build AI systems that earn their cost back. At Viettel Solutions I wrote the business case for an automated retail image-scoring product, built it, and released it — **\$500K in annual revenue**, replacing manual field-audit review. Since then I have architected a **six-service AI operations platform** now running in restaurants, and delivered **six full-stack business systems** for Vietnamese SME clients single-handedly, from requirements through security review to production deployment.

I work the way an audited organisation needs an engineer to work: **phased delivery with written sign-off gates, human-in-the-loop confirmation before data is trusted, role-based access control, complete audit trails, and rollback on every release**. All of it is independently verifiable — 797 commits across 75 peer-reviewed pull requests in production repositories.

\$500K ANNUAL REVENUE, PRODUCT SHIPPED	~3x THROUGHPUT, SAME GPU SPEND	101 SECURITY FINDINGS REMIEDIATED	6 CLIENT SYSTEMS DELIVERED SOLO	797 COMMITTS / 75 REVIEWED PRS
WHAT I DELIVERED	BUSINESS OUTCOME		VERIFIABLE EVIDENCE	
TesselAI retail image scoring — Viettel Solutions	\$500K annual revenue; manual field-audit review eliminated		Business case authored → released to production	
Sauron restaurant AI platform — DFlo	Real-time operations visibility replacing manual floor checks		399 commits, primary author	
Excel financial analytics platform	Messy client workbooks converted into governed, traceable dashboards		11 phases delivered; 101 security findings closed across 3 releases	
Phong Thanh multi-branch admin platform	One system replacing spreadsheet operations across a repair chain		21 business resources live; branch-level data segregation enforced	
Inference optimisation — TesselAI	~3x throughput on the same GPU spend — capacity without capital outlay		120 RPS at TP95 latency, no accuracy regression	

PROFESSIONAL EXPERIENCE

AI Engineer — DFlo

Jul 2025 – Present

Sole primary author of *Sauron*, the company's flagship AI platform for restaurant operations, plus three further client-facing AI products.

- Turned a manual floor-check process into a live operations view. Architected a six-service platform (API gateway, computer-vision engine, food quality control, analytics agent, dashboard, and a Postgres/MongoDB/Redis data layer) delivering real-time table occupancy, movement heatmaps, automated food quality scoring, and conversational reporting in a single product — **399 commits, primary author**.
- Removed manual quality sampling from the kitchen line. Designed an automated food quality-control pipeline using a vision-language model: classification across 24 dish variants, asymmetric scoring, and server-side automatic capture. Engineered for accountability rather than best-effort — **circuit breakers, a dead-letter queue so no evaluation is ever silently lost**, and streamed diagnostics so operators can see *why* a dish failed.
- Restored a fully failed production line and closed the gap permanently. Diagnosed a total (100%) quality-control failure caused by configuration values silently not reaching the vision model, fixed the root cause across all three consumers of that configuration, and rebuilt the retry path so exhausted retries always reach the dead-letter queue with a diagnosable reason attached.
- Unlocked multi-user operation. Built a session-isolation layer giving each operator an independent camera session — the change that took the platform from single-operator demo to concurrent commercial use.
- Made a year of footage searchable in plain language. Shipped AI video moment retrieval: managers describe what they are looking for and get the exact clip, without scrubbing recordings.
- Put reporting in the hands of non-technical managers. Built an LLM analytics agent with guardrails and conversation memory over historical operations data — questions answered without a BI request queue.
- Delivered the product in four markets from day one. Owned the customer-facing dashboard end to end with **full EN / VI / JA / ZH localisation**, accessibility remediation, and analytics panels backed by unit-tested calculation modules.
- Cut infrastructure cost without degrading the product. Reduced GPU detection cadence 30 → 5 FPS using tracker interpolation, holding visible quality while cutting GPU load; automated releases with **CI/CD and automatic rollback**.

Additional products delivered at DFlo

- **Conveyor-belt dish counting for KichiKichi (Golden Gate Group)** — production computer-vision system counting dishes by category through every stage of the belt across dual cameras, with live dashboards for store managers. Simplified the interface from 30+ endpoints to 13, materially reducing maintenance burden.
- **Enterprise video intelligence platform** — extended a research retrieval system into a production service handling **video libraries of 24+ hours**, with vector search, batch processing, and session caching for low-latency response.
- **Meeting intelligence for a distribution-centre client** — recording through transcription, AI summary, PDF/XLSX reporting, and a searchable cross-meeting knowledge base; deployed to **Microsoft Azure with infrastructure-as-code**, giving the client a reproducible, auditable environment.

Independent Full-Stack Engineer — Freelance

Nov 2024 – Present

Six production business systems delivered solo — requirements, architecture, build, security review, deployment, and handover documentation. Clients are Vietnamese SMEs with no in-house IT function, so every system had to be operable and recoverable by non-technical staff.

- **Financial analytics platform — built for people who do not trust a black box.** Clients upload inconsistent financial, KPI and HR workbooks; the platform reconciles them into structured tables, **proposes how the files relate and then waits for a human to confirm before anything is treated as fact**, and produces editable dashboards. Delivered across **11 phases with documented completion criteria**; three remediation releases closed **101 distinct security findings**, and a manual security review was completed and signed off. Accuracy gates are enforced mechanically — the test harness *refuses to report a pass* on synthetic data rather than flattering the result.
- **Multi-branch operations platform for an appliance-repair chain (94 commits)** — replaced spreadsheet operations with a single system covering **21 business resources**. Access control is structural, not cosmetic: **branch-level data segregation enforced in the access token itself**, sensitive repair actions permission-guarded, and confidential fields stripped before data leaves the server. Migrated the production host and automated database and API releases so the client's releases stopped depending on my availability.
- **Content and tax-agency systems (3 clients)** — one built deliberately in **zero-dependency Node.js** so the client can run, patch and back up the system indefinitely without a build toolchain or vendor lock-in; includes rate limiting, forced credential setup on first use, and automated search-engine indexing.
- **Commercial websites with real administrative back-ends (3 clients)** — content management, file handling, and scripted deployment, handed over with operating documentation.

AI Engineer — Rainscales (Contract)

Feb 2025 – Jun 2025

- Delivered customer and staff detection with multi-camera counting for a casino client under difficult physical camera placement — the constraint that had blocked earlier attempts.
- Built behavioural recognition (hand-raising, head-turning) so floor staff are directed to guests who need help, without additional headcount.
- Achieved **95% detection accuracy (mAP) at ~25 FPS**, meeting the client's real-time threshold on a single GPU.
- Produced warehouse layout placement rules for Linfox from depth-estimation and rack-detection analysis, converting camera output into an operational decision.

AI Engineer — Viettel Solutions

Jun 2023 – Jan 2025

TesselAI — from internal proposal to a revenue-generating product

- **Identified the opportunity, wrote the business case, then built and released the product** — automated market image scoring using computer vision, generating **\$500K in annual revenue** and replacing manual field-audit review across the retail footprint.
- **Delivered ~3x throughput on unchanged hardware** by re-engineering the inference path — 120 requests per second at TP95 latency with no loss of accuracy. Capacity gained without capital expenditure.
- Trained and deployed the detection and segmentation models running in the production scoring pipeline.
- Worked directly with business and field-operations teams to convert audit rules into measurable, testable acceptance criteria — so "correct" was defined by the business, not by the model.

GOVERNANCE, SECURITY & DELIVERY DISCIPLINE

- **Human confirmation before trust.** In the financial analytics platform, no relationship between client workbooks is acted on until a person confirms it. Automation proposes; a human approves.
- **Remediation with evidence.** 101 security findings identified and closed across three releases, plus a completed manual security review with outstanding items explicitly recorded rather than quietly dropped.
- **Access control by design.** Branch-scoped data isolation enforced at the token level, role-based permissions on sensitive actions, and least-privilege data projections.
- **No silent failure.** Dead-letter queues and diagnostic error reporting, so a failed automated evaluation is always visible and recoverable — never lost.

- **Honest test gates.** Accuracy thresholds that skip rather than pass when real validation data is unavailable, preventing false assurance.
- **Reversible releases.** Continuous delivery with automatic rollback; deployment documented as step-by-step runbooks so the client is never dependent on one individual.
- **Documented handover.** Architecture, deployment, and operating documentation maintained alongside the code as a standard, not an afterthought.

TECHNICAL CAPABILITY

Artificial Intelligence — computer vision (YOLO11/YOLOX, OpenCV, person re-identification and tracking), vision-language models (Gemini), LLM agents with guardrails (Agno, LangChain), retrieval-augmented generation, speech recognition, PyTorch, TensorFlow, ONNX, Triton

Backend & Data — Python (FastAPI, Celery, Flask), TypeScript (NestJS, Express), PostgreSQL, MongoDB, Redis, vector databases, PySpark, Pandas, Power BI

Frontend — Next.js, React, TypeScript, Tailwind, multi-locale product delivery (EN/VI/JA/ZH), automated testing

Cloud & Operations — Google Cloud, Microsoft Azure (Container Apps, infrastructure-as-code), Docker, CI/CD with automated rollback, self-hosted runners, secure reverse proxy and tunnel access

Languages — Python · TypeScript / JavaScript · SQL · Go · Bash

SELECTED PROJECTS & OPEN SOURCE

- **Building-wide CCTV action retrieval** (*team of 5*) — natural-language video search across a 100-camera network so security teams find incidents in seconds instead of hours.
- **Open-source contributor, PyTorch AnimeGAN** (*188+ stars*) — improved training stability and added multi-GPU support to a widely used image-generation project.
- **AI video clipping tool (PreenCut)** — extended an open-source retrieval tool with new capability, defect fixes, and a test suite.
- **Applied research and competition work** — NFL player-safety impact detection and helmet tracking, Google isolated sign-language recognition, medical imaging segmentation and classification.
- **Developer tooling** — a GPU memory-requirement calculator for language-model workloads, and a reinforcement-learning chess engine built from first principles.

VERIFICATION

All engineering claims are independently checkable at github.com/miniHaLe — 797 commits across 75 peer-reviewed pull requests, spanning production platforms, client systems, and open-source contributions. Commercial repositories are private; access can be arranged on request.

Languages: Vietnamese (native) · English (professional working proficiency)